

Supplementary Material:
Deployment Calibration Trace for
*The Price of Fairness, Complexity, and Incentives in
No-Numeraire Combinatorial Trade-Intent Auctions*

This anonymized supplement provides the deployment summary statistics used in Section 7 of the submission, so that the reported figures can be verified without de-anonymizing the authors. The underlying per-auction records and the reproduction scripts are included in the accompanying archive. We reiterate the scope stated in the main text: these are **observable deployment diagnostics, not estimates of the structural complementarity factor g** . The trace records winner and runner-up solution scores and order-level notionals, but not the full vector of per-leg standalone valuations required to compute $V(S)/\sum_j V(\{j\})$; the worst-case results of the submission hold for any g .

S1. Trace scale

Quantity	Value
Post-adoption solver competitions	802,816
Observation window	8 months
Covered notional	\$54.69 B
Visible accounting gap (aggregate)	\$5.13 M
Visible accounting gap (relative)	≈ 0.94 bps

Table 1: Scale of the deployment trace. The visible accounting gap is the aggregate difference between the winning solution’s score and the best non-winning visible score.

S2. Visible coverage / spread distribution

For each auction, let Score_1 be the winning solution’s score and $\text{Score}_2^{\text{other}}$ the best positive score submitted by a different solver. Define the observable

$$\hat{\alpha}^{\text{obs}} = \frac{\text{Score}_2^{\text{other}}}{\text{Score}_1} \in (0, 1], \quad \hat{G}^{\text{obs}} = 1/\hat{\alpha}^{\text{obs}} \geq 1.$$

This is an *auction-level* visible coverage; it is a proxy for, and weaker than, the per-leg coverage parameter of the submission’s coverage law, which would require per-leg standalone benchmarks.

Event	Share of auctions
Visible spread $\widehat{G}^{\text{obs}} > 2$ (all orders)	25.3%
Visible spread $\widehat{G}^{\text{obs}} > 4$ (all orders)	20.2%
Visible spread $\widehat{G}^{\text{obs}} > 2$ (small-notional orders)	63.1%

Table 2: Tail of the visible-spread distribution. A non-trivial spread is the common case rather than a corner, especially for small orders.

S3. Benchmark robustness

Under a naive local hard-reserve benchmark (reserve computed from a short local window with no inheritance or smoothing), we replay a single low-quote injection per cell-window and record whether it would move the published benchmark enough to admit an otherwise-filtered settlement.

Benchmark variant	Vulnerable cell-windows (auction-weighted)
Local hard reserve (naive)	97.1%

Table 3: Fragility of a naive benchmark to single-injection contamination, motivating the delayed / quantile / rate-limited / parent-cell-inherited benchmarks analyzed in the submission’s contamination game. The replay parameters and robust-reserve configuration are in the accompanying scripts.

S4. Reproduction

The accompanying archive contains: (i) the processed per-auction records underlying Tables S1–S3; (ii) the scripts that compute the visible coverage distribution and the accounting gap; and (iii) the shadow-reserve replay used for the robustness diagnostic, with its configuration file. All identifying metadata (author, affiliation, project, and protocol-internal identifiers beyond what is needed to interpret the tables) has been removed.